

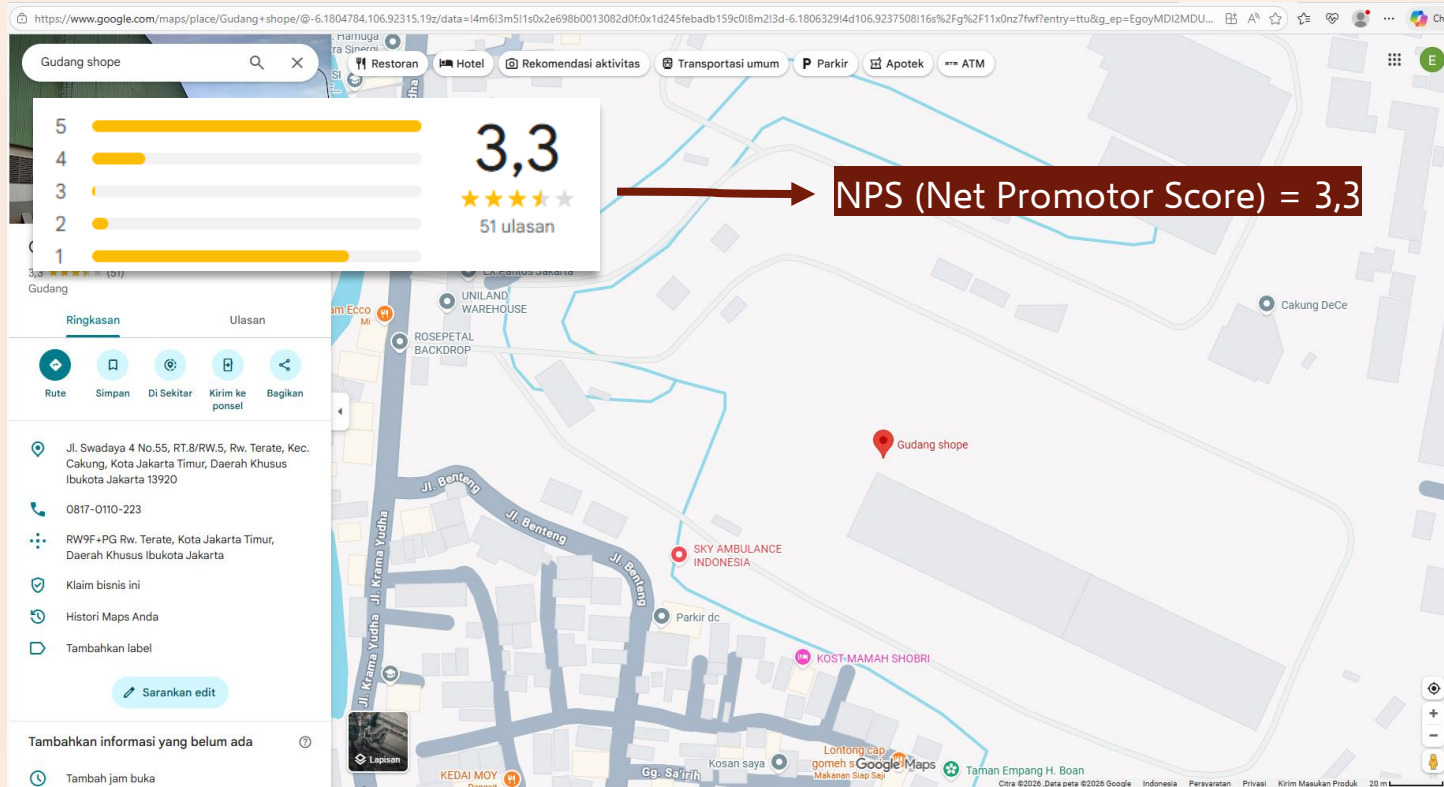
REDUCING LOST AND STUCK PARCELS THROUGH AI SORTING MONITORING OPTIMATION Using computer vision

Shopee Express DC Cakung study case
by Emmy Jacklyn Pontoan

How might we apply AI-driven automation to optimize workflows and increase operational efficiency in logistics?

Shopee Express DC Cakung study case

GOOGLE MAPS REVIEW



Warehouse Shopee DC Cakung, Jaktim

8/8 = 100% Lost/stuck inside warehouse

- Tretan Wahbrain**
Local Guide · 5 ulasan · 20 foto

★★★★★ 4 minggu lalu

PENGIRIMAN LAMBAT, SUDAH LEBIH DARI 10 HARI BARANG TIDAK DATANG

Suka Bagikan
- IMAM ALFIAN SCP**
1 ulasan

★★★★★ sebulan lalu

Paket dr tgl 26 smp tgl 29 masih stuck di gudang cakung. Sumpah smp capek laporan yang jawaban nya gitu2 doang. Makin ancur pelayanan spx gak jelas

Suka Bagikan
- Shandy Abdul Azis**
Local Guide · 18 ulasan · 14 foto

★★★★★ 6 bulan lalu

Tutup aja cabang gudang ini ga guna masa paket gw dari tanggal 5 sampe tanggal 8 sekarang masih di gudang,sampah sampah

1 Suka Bagikan
- rizqi syarif**
5 ulasan · 14 foto

★★★★★ 5 bulan lalu

Paket dari tgl 5 des estimasi tgl 9 tapi skrg tgl 19 masih di dc, gak ada kejelasan sama sekali, PAYAH

Suka Bagikan
- Abduel Raymon**
1 ulasan

★★★★★ 5 bulan lalu

Pakett gua hilang

Suka Bagikan
- Anisa Ica**
1 ulasan

★★★★★ 5 bulan lalu

Paling gak jelas ini gudang. Paket sebulan ngendap gak dateng2

Suka Bagikan
- artika maret**
4 ulasan · 1 foto

★★★★★ 2 bulan lalu

Paket saya stuck disini. Ktanya salah sortir penjaluran paket. Tp sdh 24 jam lbh pkt stuck di gdg ini gak gerak2

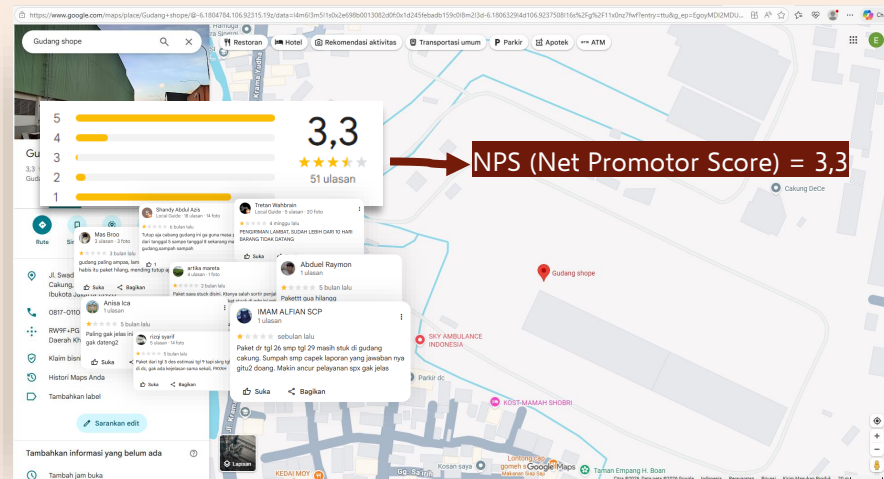
Suka Bagikan
- Mas Broo**
3 ulasan · 3 foto

★★★★★ 3 bulan lalu

gudang paling ampas, lama amat nyortir paket, trus habis itu paket hilang, mending tutup aja ini cabang

Suka Bagikan

GOOGLE MAPS REVIEW



Warehouse Shopee DC Cakung, Jaktim

M, Cakung DC spv spx worker

“..biasanya waktu proses inbound atau outbound orang yang ngangkut itu ga bener jadinya tali tisnya putus, barang kececer dari karung, nah itu udah asal lempar aja ke troli, ga tau jatuh darimana..”

6/6 = 100% Lost/stuck inside warehouse

Main problem:

Parcels scattered during manual sorting process, mis-routed, stuck parcels inside warehouse



B, Cakung DC dw spx worker

“..tiap hari banyak banget paket yang bermasalah, dari 800-900 yang perlu diproses, sekitar 100 pasti ada aja yang bermasalah..”

R, Cakung DC EH/Tracer spx worker

“..banyak barang yang waktu proses unpacking itu barang tercecer, lama ngendep, pas nemu udah ga bisa diproses, ujungnya jadi barang sampah..”

V, Cakung DC EH/Tracer spx worker

“..banyak juga paket yang jatuh, nyelip di kolong..”

S, Cakung DC EH/Tracer spx worker

“..selalu ada aja barang yang ke bagian EH/Tracer, kalau ga ada barcode bisa dicari, tapi kalau barangnya kurang harus dicari dulu baru bisa diproses, misal harusnya 10 tapi adanya 5..”

H, Cakung lead hub spx worker

“.. paket sering salah sort, salah wilayah, mau ga mau harus diretur...,kasian ke kurirnya mba kalau banyak paket yang retur, gajinya mereka aja hanya 2rb per paket, jadinya berkurang pendapatan mereka per hari..”



Main problem:

Parcels scattered during manual sorting process, mis-routed, stuck parcels inside warehouse



Time consuming manual investigation

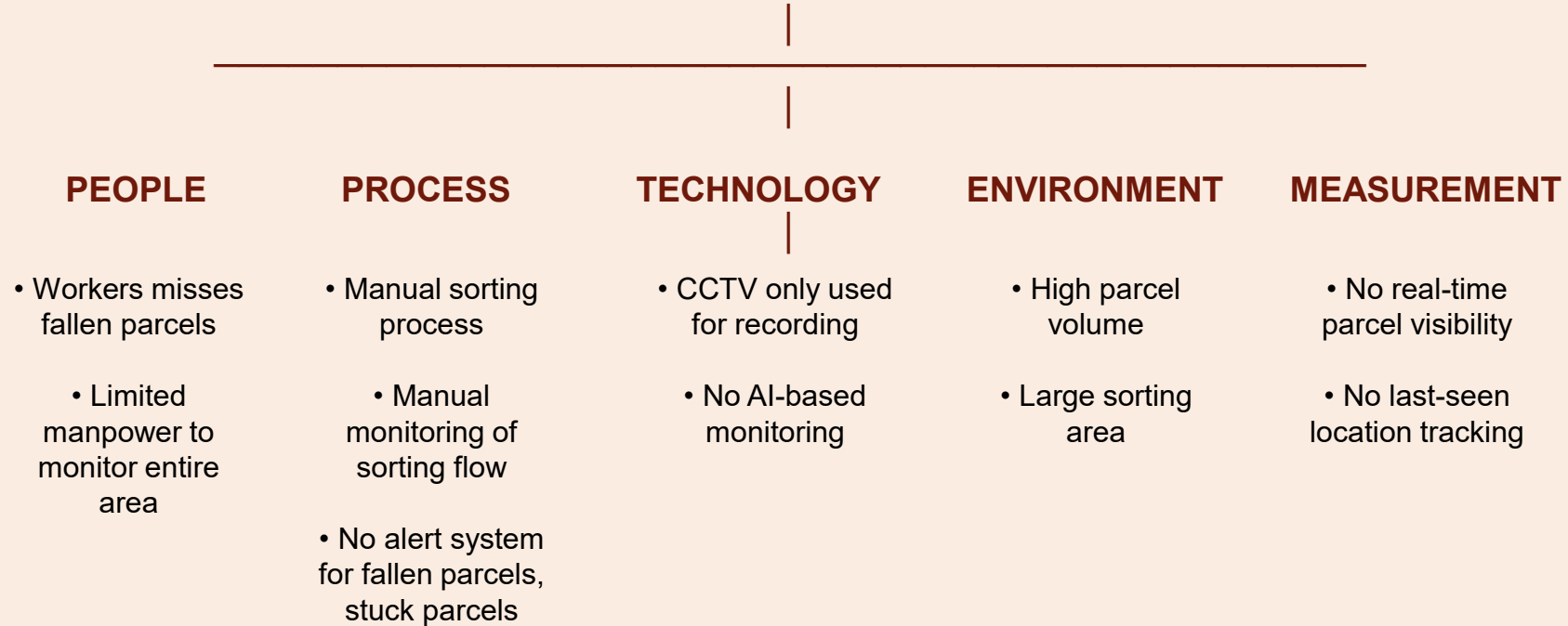


Possible refunds, vouchers for late arrival



causing operational inefficiency

LOST & STUCK PARCELS



Root cause:

Lack of real-time monitoring throughout the sorting process.

How might we apply AI-driven automation to optimize workflows and increase operational efficiency in logistics?

Shopee Express DC Cakung study case

How might we apply AI-driven automation to optimize workflows and increase operational efficiency in logistics?

**AI Sorting Monitoring
system using
Computer Vision**

As-Is Process

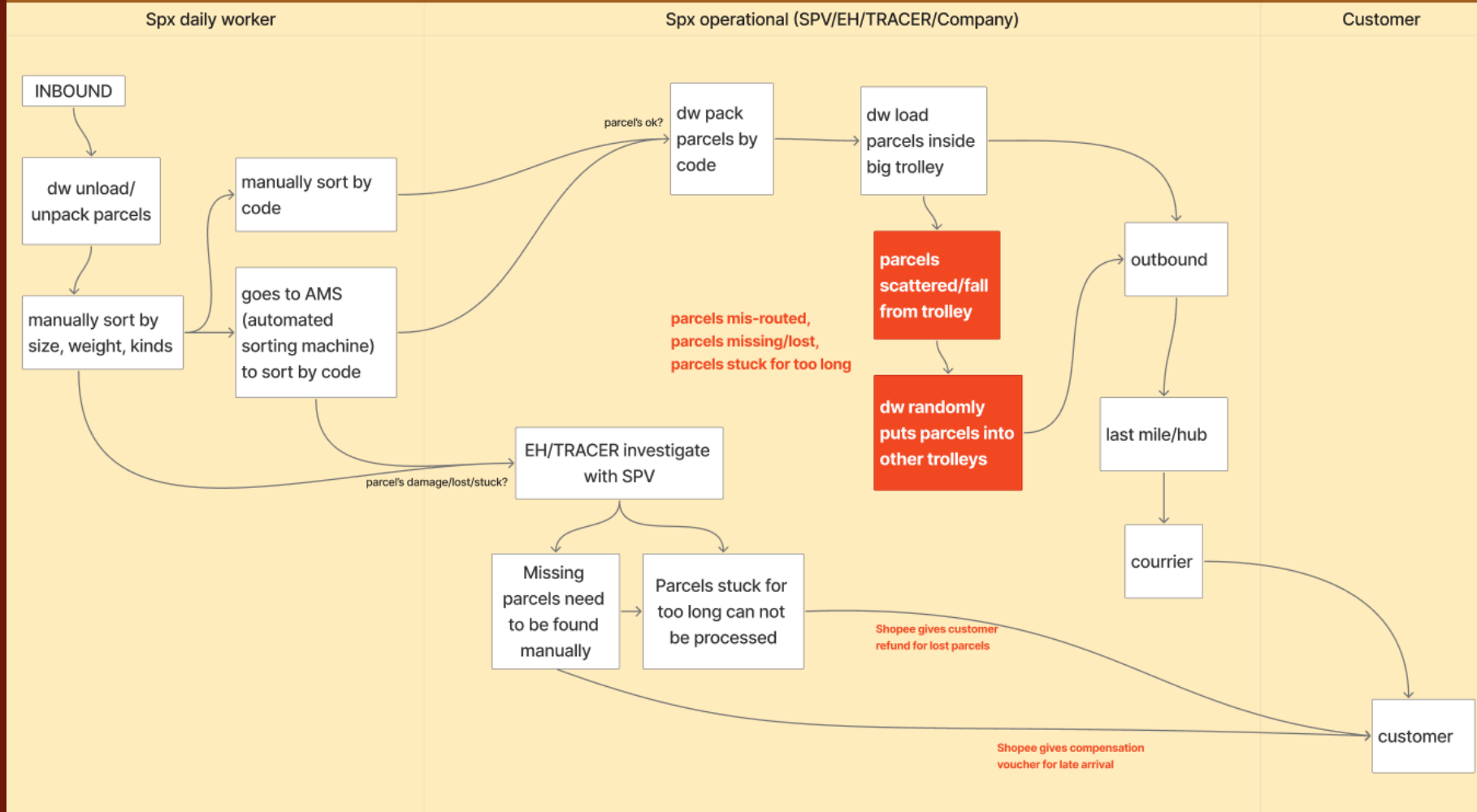
Inbound → Manual Sorting process → Parcels fall and stuck/mis-routed → Outbound → **Return parcels** → **Customer complain** → **Compensation**

To-Be Process

Inbound → **AI Monitoring CCTV** Sorting process → Real-time detection alert for fallen/stuck/mis-routed parcels → SPV Action → Outbound

As-Is Process:

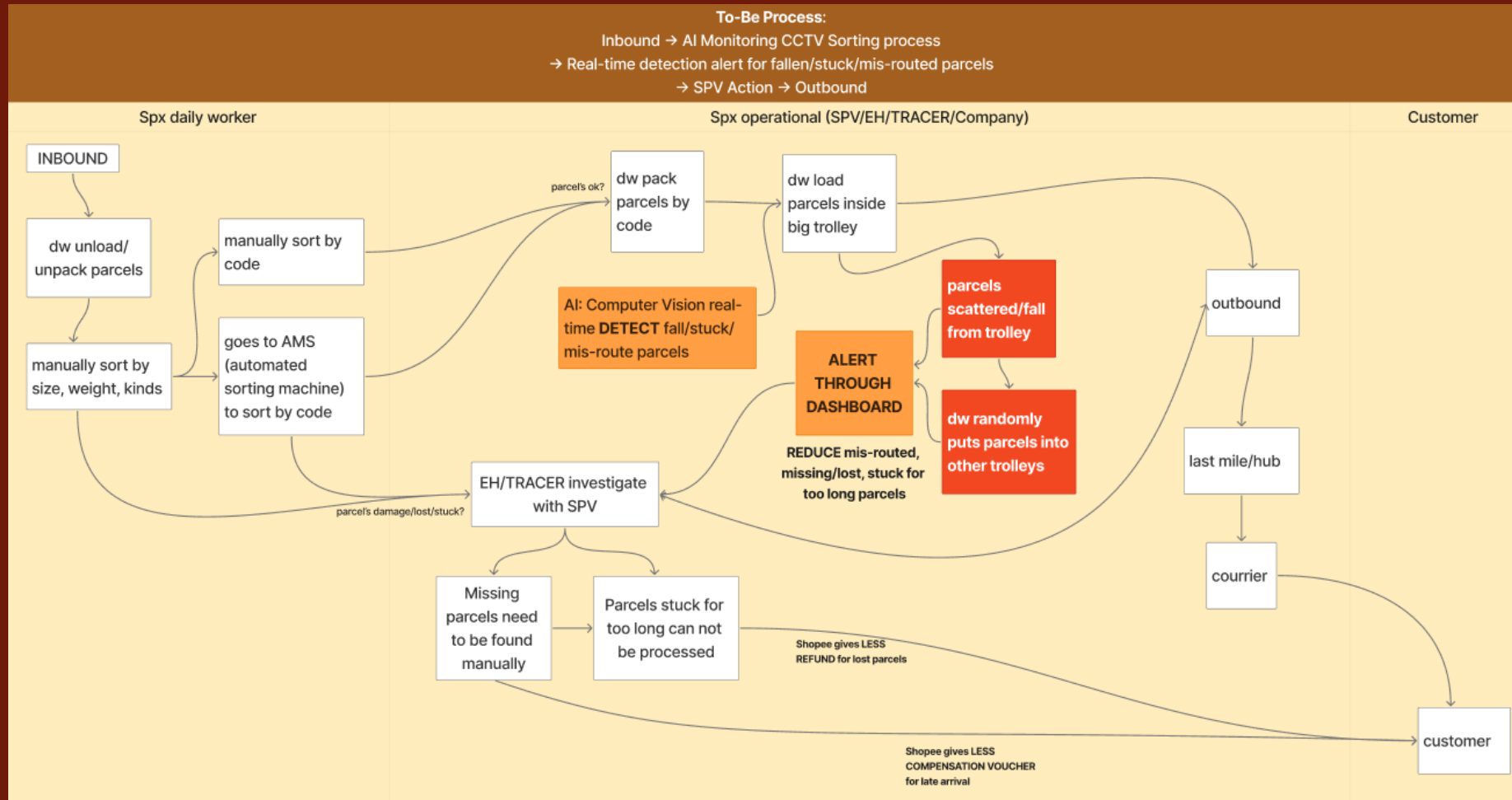
Inbound → Manual Sorting process → Parcels fall and stuck/mis-routed
→ Outbound → Return parcels → Customer complain → Compensation



1. Increased customer complaints for lost and delayed (stuck) parcels

2. Operations doing time-consuming investigation manually

3. Higher compensation costs



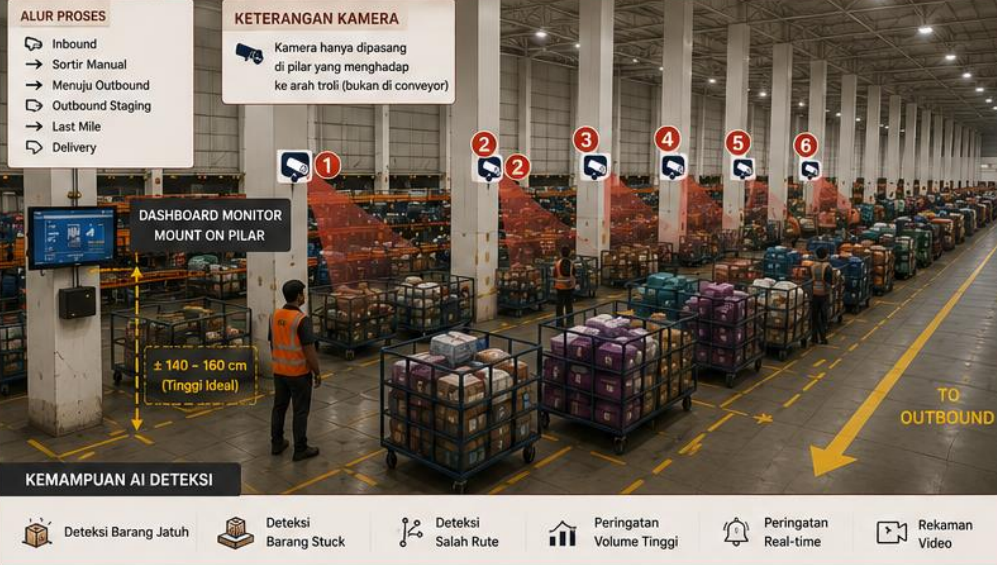
1. **REDUCED CUSTOMER COMPLAINTS** for lost and delayed (stuck) parcels

2. Operations **EFFICIENTLY SAVES TIME** prevent problems

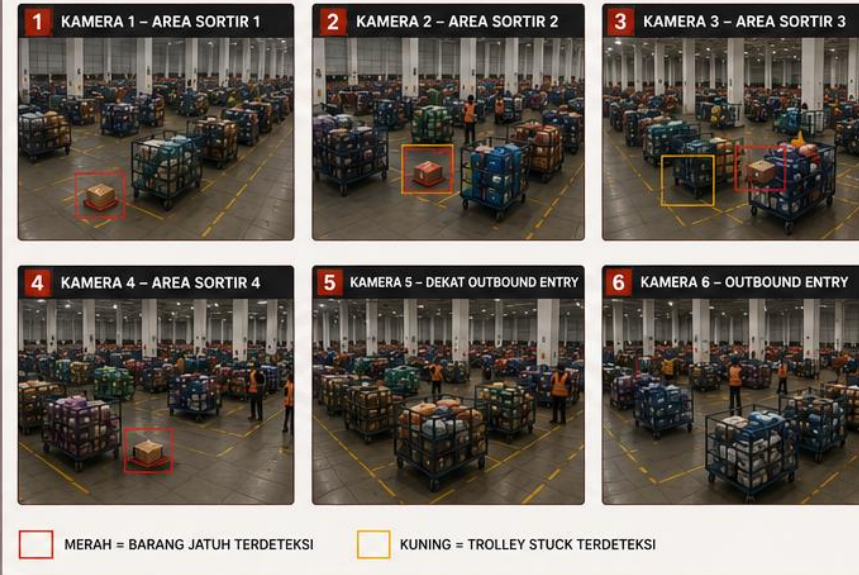
3. **LOWER COMPENSATION COSTS**

SIMULASI AI MONITORING – 6 KAMERA (FOKUS PADA TROLLEY)

Deteksi Barang Jatuh & Stuck di Area Sortir Manual Menuju Outbound



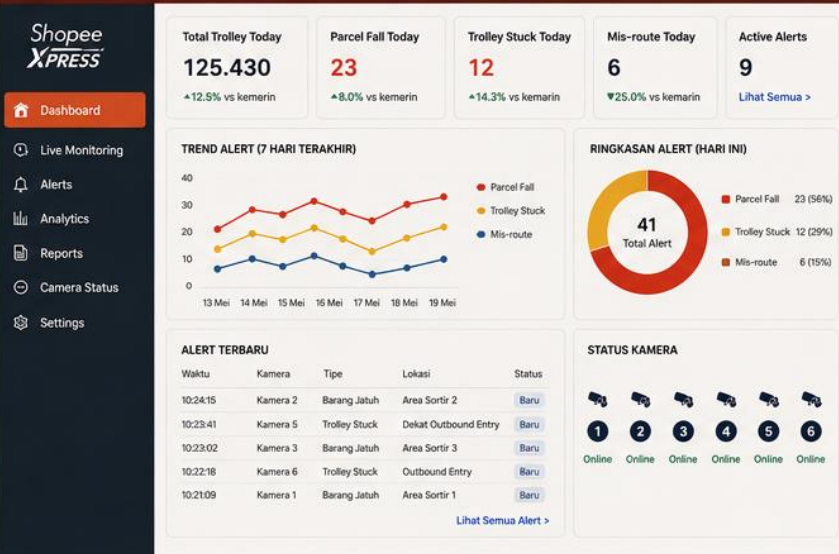
6 KAMERA – REAL TIME VIEW (FOKUS PADA TROLLEY)



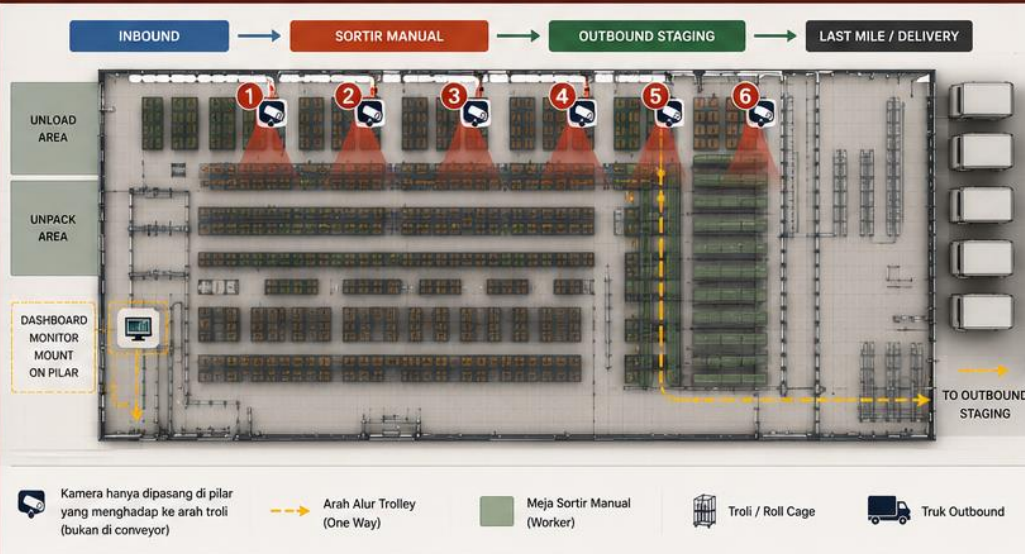
PILOT PROJECT – AI MONITORING SIMULATION

- 6 CCTV Cameras integrated with AI
- Dashboard monitor & report
- Focus on sort to outbound area

AI MONITORING DASHBOARD (CONTOH)



BLUEPRINT PELETAKAN KAMERA (FOKUS PADA TROLLEY)



As-Is Process (Reactive)

Inbound → Manual Sorting process → Parcels fall and stuck/mis-routed → Outbound → **Return parcels** → **Customer complain** → **Compensation**

To-Be Process (Proactive)

Inbound → **AI Monitoring CCTV** Sorting process → Real-time detection alert for fallen/stuck/mis-routed parcels → SPV Action → Outbound

KPI & OKR

Key Result (OKR)	KPI	Baseline	Target
Reduce lost parcels during sorting operations	Lost Parcel Rate	Current	↓ 50%
Reduce parcels stuck within the warehouse sorting process	Stuck Parcel Rate	Current	↓ 50%
Reduce manual investigations and exception handling workload	EH/Tracer Case Rate	Current	↓ 40%
Reduce parcel investigation time	Parcel Recovery Time	Current	↓ 60%

Objective:

Improve warehouse sorting visibility and reduce lost & stuck parcels through AI-powered monitoring.

RISK ASSESSMENT & MITIGATION

Risk	Potential Impact	Mitigation
AI fails to detect all fallen parcels	Some parcels may still become lost	Conduct a pilot project and continuously retrain the AI model using actual warehouse data
Too many false alerts	Operators may ignore notifications and reduce trust in the system	Adjust detection thresholds and validate alerts during the pilot phase
Camera Blind Spots	Fallen or misplaced parcels may not be captured by the system	Optimize camera placement with warehouse mapping and perform regular coverage assessments

(ASSUMPTION) COST STRUCTURE – PILOT PROJECT

Component	Cost
AI Vision Service	≈Rp118.800.000
Storage	≈Rp2.400.000
Dashboard	≈Rp25.000.000
Integration	≈Rp20.000.000
Training	≈Rp10.000.000
Contingency	≈Rp10.000.000
Total	≈Rp186.200.000

AI Vision service

$6 \text{ Camera} \times \text{USD}100 \times 12 \text{ months} = \text{USD } 7.200$
 Kurs Rp16.500
 ≈ **Rp118.800.000/year**

Sorting area to outbound

System Integration

≈ **Rp20.000.000/year**

Cloud Storage

$100 \text{ GB/Camera} \times 6 \text{ Camera} = 600 \text{ GB}$

Training

≈ **Rp10.000.000/year**

Dashboard Development

≈ **Rp25.000.000/year**

Contingency

≈ **Rp10.000.000/year**

Google Storage:

USD0.02/GB

600×0.02

= USD12/month

≈ Rp198.000/month

≈ **Rp2.400.000/year**

Source:

[Google Cloud Vision AI Pricing](#)

ROI & BEP – PILOT PROJECT (ASSUMPTION)

Investasi Awal (Year 1)

Total Investment ≈ Rp186.200.000

Benefit

Reducing Lost/Stuck Parcel ≈ Rp90.000.000

Total Benefit ≈ Rp90.000.000/year

Year 1

Investment ≈ Rp186.200.000

Benefit ≈ Rp90.000.000

Year 2

Additional Benefit ≈ Rp90.000.000

Accumulated Benefit ≈ 180 million

Year 3

Additional Benefit ≈ Rp90.000.000

Accumulated Benefit ≈ 270 million

BEP ≈ bulan ke-25
(Q1 Tahun Ketiga)



Perhitungan Payback Period

Rumus:

$$\begin{aligned} \text{Payback Period} &= \frac{\text{Investment}}{\text{Annual Benefit}} \\ &= \frac{186.200.000}{90.000.000} \\ &= 2,07 \text{ year} \end{aligned}$$

or

$$2,07, \times 12 = 24,8 \text{ months}$$

≈ 25 months

Thank you.